

LINEAR ALGEBRA AND CALCULUS

(CIVIL, EEE, MECH, ECE, CSE, IT, AIDS, AIML)

Time: 3 hours

Max Marks: 70M

- Note:** 1. Question paper consists of two parts (**Part A and Part B**)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART – A				10 X 2 = 20 M		
			Marks	BTL	CO	
I.	a)	Find the value of k such that the rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is 2.	2M	L1	CO1	
	b)	Define diagonal dominance property.	2M	L1	CO1	
	c)	Find the Characteristic polynomial of $A=\begin{bmatrix} 2 & 5 \\ 1 & -3 \end{bmatrix}$.	2M	L1	CO2	
	d)	State the Cayley-Hamilton theorem.	2M	L2	CO2	
	e)	State Cauchy’s mean value theorem.	2M	L2	CO3	
	f)	Expand $\sin x$ in powers of ‘x’.	2M	L2	CO3	
	g)	Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $z = x^2y - x\sin(xy)$.	2M	L3	CO4	
	h)	If $x = u(1 + v), y = v(1 + u)$ then show that $\frac{\partial(x,y)}{\partial(u,v)} = 1 + u + v$.	2M	L2	CO4	
	i)	Evaluate $\int_1^2 \int_1^3 xy \, dx \, dy$	2M	L5	CO5	
	j)	Evaluate $\int_0^1 \int_1^2 \int_1^3 xyz \, dx \, dy \, dz$	2M	L5	CO5	
PART – B				5 X 10 = 50 M		
UNIT – I						
1.	a)	Determine the rank of the matrix A by reducing it to the normal form where $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$	5M	L5	CO1	
	b)	Solve completely the system of equations: $x + y + w = 0$; $y + z = 0$; $x + y + z + w = 0$; $x + y + 2z = 0$.	5M	L3	CO1	
(OR)						
2.	a)	Solve the equations $2x_1 + x_2 + x_3 = 10$; $3x_1 + 2x_2 + 3x_3 = 18$; $x_1 + 4x_2 + 9x_3 = 16$ using Gauss-Elimination method	5M	L3	CO1	
	b)	Using Gauss-Seidel iteration method to solve the system $10x + y + z = 12$; $2x + 10y + z = 13$; $2x + 2y + 10z = 14$.	5M	L3	CO1	
UNIT – II						
3.		Verify Cayley-Hamilton theorem and also find A^4 if $A = \begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$	10M	L5	CO2	
(OR)						
4.		Reduce the quadratic form $3x^2 + 5y^2 + 3z^2 - 2xy + 2zx - 2yz$ to canonical form by orthogonal reduction and also find index, signature and nature of the quadratic form.	10M	L4	CO2	

UNIT – III					
5.	a)	Verify Rolle's theorem for $f(x) = (x - a)^m(x - b)^n$ where m,n are positive integers in $[a, b]$.	5M	L3	CO3
	b)	Verify Cauchy's mean value theorem for $f(x) = \sin x$ and $g(x) = \cos x$ in $[a, b]$.	5M	L3	CO3
(OR)					
6.	a)	Using Taylor's theorem express the polynomial $2x^3 + 7x^2 + x - 6$ in powers of $(x - 1)$.	5M	L3	CO3
	b)	Expand e^x in powers of x	5M	L3	CO3
UNIT – IV					
7.	a)	If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$.	5M	L5	CO4
	b)	Expand $x^2y + 3y - 2$ in powers of $(x - 1)$ and $(y + 2)$.	5M	L3	CO4
(OR)					
8.	a)	Verify whether the following functions are functionally dependent, if so, find the relation between them $u = \frac{x+y}{1-xy}$ $v = \tan^{-1} x + \tan^{-1} y$.	5M	L3	CO4
	b)	Apply Lagrange's method to find a point on the plane $3x + 2y + z - 12 = 0$ which is nearest to the origin.	5M	L3	CO4
UNIT – V					
9.		Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy \, dx dy$ and hence evaluate.	10M	L5	CO5
(OR)					
10.	a)	Evaluate $\iint r^3 dr d\theta$ over the area bounded by the circles $r = 2\sin\theta$ and $r = 4\sin\theta$.	5M	L5	CO5
	b)	Evaluate $\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dz dx dy$.	5M	L5	CO5

*** End of the Question Paper ***